

1. Initial Analysis & Troubleshooting

Initial standard analysis using tools like `exiftool` and `zsteg` revealed a corrupted PNG file signature:

```
[!] #<ZPNG::NotSupported: Unsupported header "\x89PNG\r\n\x1A\xA0" in #<File:chall.png>>
```

The standard magic bytes for a valid PNG file are `\x89\x50\x4e\x47\x0d\x0a\x1a\x0a`, but the target file had `\xA0` instead of `\x0A` at the end of its 8-byte header structure.

2. Exploitation & Fix

To fix the file signature corruption directly from the Kali terminal, `dd` was used to overwrite the corrupted bytes with a valid PNG magic header structure:

```
Bash
printf '\x89\x50\x4e\x47\x0d\x0a\x1a\x0a' | dd of=chall.png bs=1 count=8 conv=notrunc
```

3. Steganography Extraction

Once the PNG format structure was repaired, `zsteg` successfully parsed the image chunks and analyzed the Least Significant Bits (LSB).

```
Bash
zsteg -a chall.png
```

Result:

The tool discovered plain text hidden inside the `b1,rgb,lsb,xy` data stream:

```
Plaintext
b1,rgb,lsb,xy    .. text: "33:CYBERNAUTS{Y0u_g07_7h3_fr33_fl4g}"
```

Flag

`CYBERNAUTS{Y0u_g07_7h3_fr33_fl4g}`

Session Actions Edit View Help

(base) └─(kali㉿kali)-[~/Desktop/NSU]

└─\$ strings chall.png | grep -i "cybernavts"

(base) └─(kali㉿kali)-[~/Desktop/NSU]

└─\$ zsteg -a chall.png

```
b1,b,lsb,xy      .. file: OpenPGP Public Key
b1,rgb,lsb,xy    .. text: "33:CYBERNAUTS{Y0u_g07_7h3_fr33_fl4g}"
b1,bgr,lsb,xy    .. text: "BQY#AUQE"
b1,rgba,lsb,xy   .. text: "S[3=Q[[YY"
b3,abgr,msb,xy   .. file: MPEG ADTS, layer I, v2, 256 kbps, Monaural
b5,abgr,lsb,xy   .. file: MPEG ADTS, layer II, v1, 384 kbps, Monaural
b5p,abgr,lsb,xy  .. file: AIX core file 64-bit
b6,b,msb,xy      .. file: MPEG ADTS, layer I, v2, Monaural
b6,bgr,msb,xy    .. file: MPEG ADTS, layer I, v2, 112 kbps, Monaural
b8,g,msb,xy      .. file: RDI Acoustic Doppler Current Profiler (ADCP)
b8,rgb,msb,xy    .. file: RDI Acoustic Doppler Current Profiler (ADCP)
b8,rgba,msb,xy   .. file: RDI Acoustic Doppler Current Profiler (ADCP)
b8,abgr,msb,xy   .. file: ddis/ddif
b3,rgb,msb,xy,prime .. file: MPEG ADTS, layer I, v2, 112 kbps, Monaural
b4,r,lsb,xy,prime .. file: AIX core file 64-bit
b4,r,msb,xy,prime .. file: RDI Acoustic Doppler Current Profiler (ADCP)
b4,g,msb,xy,prime .. file: MPEG ADTS, layer I, v2, 112 kbps, 24 kHz, JntStereo
b4,rgba,msb,xy,prime .. file: MPEG ADTS, layer I, v2, 24 kHz, Monaural
b5,r,msb,xy,prime .. file: MPEG ADTS, layer II, v1, 48 kHz, Monaural
b5,g,lsb,xy,prime .. file: MPEG ADTS, layer II, v1, JntStereo
b5,rgb,lsb,xy,prime .. file: MPEG ADTS, layer II, v1, 384 kbps, Monaural
b5,rgba,lsb,xy,prime .. file: MPEG ADTS, layer II, v1, JntStereo
b5,abgr,msb,xy,prime .. file: MPEG ADTS, layer II, v1, Monaural
b5p,r,lsb,xy,prime .. file: AIX core file 64-bit
b5p,r,msb,xy,prime .. file: RDI Acoustic Doppler Current Profiler (ADCP)
```